

Task	Prompt	Answer
bipartite maximum matching	The task is to determine the maximal matching in a bipartite graph. The input graph is guaranteed to be a bipartite graph. Here is an undirected graph containing nodes from 1 to 4. The edges are: (1, 3), (1, 4), (2, 3), (2, 4). Question: What is the bipartite maximal matching of the bipartite graph? You need to format your answer as a list of edges in ascending dictionary order, e.g., [(u1, v1), (u2, v2), ..., (un, vn)].	[(1, 3), (2, 4)]
bridges	The task is to find all bridges of a graph. A bridge is an edge in a graph whose removal increases the number of connected components. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 5. The edges are: (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5). Question: What are the bridges of the graph? You need to format your answer as a list of edges in ascending dictionary order, e.g., [(u1, v1), (u2, v2), ..., (un, vn)].	[]
center	The task is to determine the center of a graph. The center of a graph includes the node that minimizes the maximum distance to any other nodes in the graph. The input graph is guaranteed to be connected. Here is an undirected graph containing nodes from 1 to 6. The edges are: (1, 5), (5, 2), (2, 6), (6, 4), (3, 4). Question: What is the center of the graph? You need to format your answer as a list of nodes in ascending order, e.g., [node-1, node-2, ..., node-n].	[2, 6]
closeness centrality	The task is to determine the closeness centrality of a node in the graph. For a node *u*, closeness centrality is the reciprocal of the average shortest path distance to *u* over all *n-1* reachable nodes. For directed graphs, it computes the incoming distance to *u*. Here is an undirected graph containing nodes from 1 to 8. The edges are: (1, 3), (3, 6), (2, 8), (2, 6), (8, 6), (8, 7), (4, 7), (7, 5). Question: What is the closeness centrality of node 2 in the graph? You need to format your answer as a float number.	0.4667
clustering coefficient	The task is to compute the clustering coefficient for a given node. For unweighted graphs, the clustering of a node is the fraction of possible triangles through that node that exist. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 4), (1, 5), (1, 3), (4, 2), (4, 3), (4, 5), (4, 6), (4, 7), (5, 2), (5, 3), (5, 6), (5, 7), (2, 6), (2, 7), (6, 7). Question: What is the clustering coefficient of node 6? You need to format your answer as a float number.	1.0
common neighbor	The task is to determine common neighbors between two nodes in the graph. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 7), (1, 6), (1, 4), (1, 5), (7, 2), (7, 3), (6, 2), (4, 3), (5, 3). Question: What are the common neighbors between node 2 and node 3? You need to format your answer as a list of nodes in ascending order, e.g., [node-1, node-2, ..., node-n].	[7]